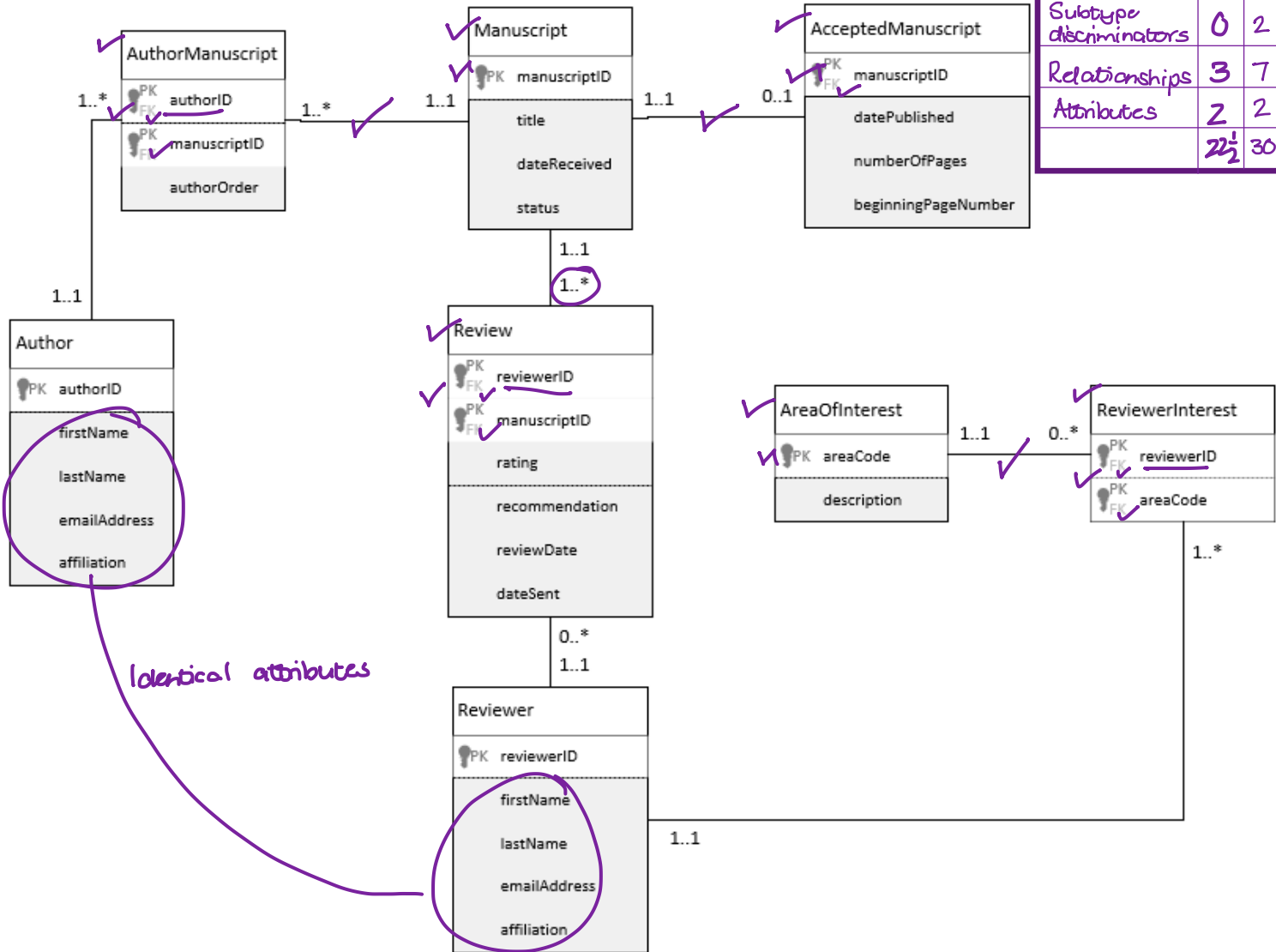


40¹/₂
51

[22¹/₂]

Modelling and Normalisation

1. Database design.



	MARK	MAX
Entities	6	7
PK	1 ¹ / ₂	5
FK	7	7
Subtype discriminators	0	2
Relationships	3	7
Attributes	2	2
	22 ¹ / ₂	30

2.

a) CarHire (carReg[pk][fk], cNo[pk][fk], make, cName, cPhone, hireDate, outletNo, outletLoc) ✓ 1

b) **Desirable dependencies:**

- carReg, cNo → make, cName, cPhone, hireDate, outletNo, outletLoc ✓✓

Undesirable dependencies

Transitive dependency:

- outletNo → outletLoc ✓✓

Partial dependency:

- carReg → make, outletNo, outletLoc No ✓

Assumption: they say a car is registered at a specific outlet, meaning if we have a carReg, we can also know where that car is located. Check data given 7

- cNo → cName, cPhone ✓✓

c) Normalisation

- CAR(carReg[pk], make, ~~outletNo[fk]~~)

Assumption: to eliminate the transitive dependency from the CAR table, we eliminated the dependent outletLoc, created another table (OUTLET) and linked both tables by outletNo.

- CUSTOMER(cNo[pk], cName, cPhone) ✓
- CAR HIRED(carReg[pk][fk], cNo[pk][fk], hireDate). Missing attribute
- OUTLET(outletNo[pk], outletLoc) ✓ 3

3. The dependency diagram is in the First Normal Form(1NF), since the NMU data provided is arranged in a **table format**, with **no repeating groups**, as well as an **established composite primary key (PK)** of barCode, empNo. ✓✓ 7

Dependencies are as follows:

Desired dependency: (barCode, empNo → itemDesc, rmNo, bldgNo, bldName, mgrName)

Undesired dependencies

Partial dependencies: - (barCode → itemDesc, rmNo) ✓✓

- (empNo → mgrName) ✓✓

Transitive dependency: (bldgNo → bldName) ✓✓ 7